

Lesson 18 — A Sequence That Solves a Problem

Matches LEGO Teacher Portal lesson: "Deep Sea Directions".

Objective (I can...): I can build a sequence that gets a real little job done. **CS concept:** Programming a sequence to solve a problem · **Builds on:** Lesson 17 (fixing and developing sequences).

AI Connection (2 min, read aloud): A robot vacuum follows a sequence to clean a room — go, turn, go, turn — until the job is done. *Ask the group:* "What's a little job you'd love a robot to finish for you?"

Materials: LEGO CS & AI kit + device per pair, a simple "problem card" (e.g., move a block from here to there, or guard a spot). **Model reuse:** Sequences model from Lesson 16 — code only.

60-minute flow - Warm-up (10): "What's the job?" — coach names a little problem (the toy needs to get to the box); pairs say the steps out loud to solve it. - **Concept + demo (8):** Pick a problem card; coach decomposes it into an ordered sequence and codes it live, running to check it solves the job. - **Build/Code (25):** Pairs pick a problem card and code a sequence that solves it on the built model (move the block, deliver the mail, do the dance). They test and adjust until the job is done. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Solve the same problem a second way with different steps; which is simpler? - **Share + tidy-up (10):** Pairs show their machine solving the job and name the key step. Model stays built.

Pair roles: Co-plan; Driver codes / Navigator checks each step solves part of the job; swap at the wiggle break. (*Trio rotates plan/code/test.*)

Differentiation: *Support* — a problem card with the first two steps given. *Extension / fast-finisher* — a problem that needs 6+ steps. *Reteach* — act out solving the problem with bodies first, then map to blocks.

Exit check: Pair's sequence actually solves the problem on the card. **At-home (optional):** Pick a little problem at home (socks all over); write the steps to solve it. ~5 min.
